AP-Mamba: Anatomy-Aware State Space Models for Multi-organ Abdominal MRI Segmentation

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***Abstract*— Accurate multi-organ abdominal MRI segmentation remains challenging due to anatomical variability, low-contrast boundaries, and complex organ co-occurrence. We propose AP-Mamba, an anatomy-aware state space model that integrates selective state space models with explicit anatomical priors. A lightweight Transformer-based Anatomical Prior Module captures global organ layout and co-occurrence cues, while a structure-sensitive decoder with attention-gated skip connections suppresses background interference and refines boundaries. The framework performs slice-wise inference and reconstructs 3D volumes for case-level evaluation. Experiments on CHAOS-MRI (5-fold cross-validation) and AMOS-MRI (held-out test set) show consistent gains over strong CNN-, Transformer-, and recent Mamba-based baselines in Dice similarity coefficient and normalized surface distance. Ablation studies verify the complementary effects of APM and attention gating, highlighting the effectiveness of explicit anatomical priors for robust multi-organ MRI segmentation.**

I. INTRODUCTION

Magnetic Resonance Imaging (MRI) provides excellent soft-tissue contrast and is widely used for abdominal diagnosis and treatment planning. Reliable multi-organ segmentation is fundamental to downstream tasks such as volumetry and surgical planning, but remains difficult due to low inter-organ contrast, indistinct boundaries, and large inter-subject anatomical variability.

CNN-based methods learn strong local representations but are limited in capturing long-range context. Transformers improve global modeling via self-attention, yet their quadratic complexity hampers high-resolution deployment [1]. Recently, State Space Models (SSMs) such as Mamba offer linear-complexity long-range modeling [2], [3].

Nevertheless, most Mamba-based segmentation approaches aggregate context implicitly and do not explicitly encode anatomical priors. For abdominal MRI, organs follow relatively stable spatial layouts and co-occurrence patterns; ignoring these constraints can yield anatomically implausible predictions (e.g., false positives and fragmented boundaries), especially under low-contrast conditions [3].

To address this, we propose AP-Mamba, an anatomy-aware framework for multi-organ MRI segmentation. Our contributions are:

- **Anatomy-aware VSSM framework:** A U-shaped architecture that integrates explicit anatomical priors into an efficient VSSM backbone [2].
- **Dual structural refinement:** An Anatomical Prior Module (APM) captures global organ layout at the bottleneck, and attention-gated skip connections refine boundaries while suppressing background responses.
- **Strong empirical results:** Consistent improvements over competitive CNN-, Transformer-, and recent Mamba-based baselines on abdominal MRI benchmarks.

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II. METHODS

As illustrated in Fig. 1, AP-Mamba follows a U-shaped encoder–decoder design. The encoder adopts stacked Visual State Space blocks (VSSM) to efficiently learn hierarchical representations [2], [3]. At the bottleneck, an Anatomical Prior Module (APM) based on a lightweight ViT injects global anatomical context to enhance long-range spatial modeling [1], [4]. The decoder uses Attention Gates (AGs) on skip connections to adaptively filter low-level features with high-level cues, suppressing background responses and refining organ boundaries during upsampling.

A. Mamba-based Visual State Space Module (VSSM)

As shown in Fig. 2, the AP-Mamba encoder adopts Visual State Space (VSS) blocks to model long-range dependencies with linear complexity [2], [3]. Following prior VSSM designs, selective state-space modeling is extended to 2-D feature maps via a multi-directional scan-and-merge operator [3], [5].

Selective state-space operator. Given a 1-D sequence $\{x_k\}$, the discretized state-space model updates

$$h_k = \bar{\mathbf{A}} h_{k-1} + \bar{\mathbf{B}} x_k, \qquad y_k = \mathbf{C} h_k. \tag{1}$$

The selective variant makes the dynamics input-dependent for content-adaptive filtering [2]. For 2-D features, we apply it along multiple directions and merge the results, denoted as $\mathrm{SS2D}(\cdot)$.

VSS block. For $U \in \mathbb{R}^{H\times W\times C}$,

$$[\mathbf{x}, \mathbf{z}] = \mathrm{LN}(U)\mathbf{W}_1, \tag{2}$$

$$\tilde{\mathbf{x}} = \mathrm{SS2D}(\mathrm{SiLU}(\mathrm{DWConv}(\mathbf{x}))), \tag{3}$$

$$Y = U + (\tilde{\mathbf{x}} \odot \mathrm{SiLU}(\mathbf{z}))\,\mathbf{W}_2, \tag{4}$$

where $\mathbf{W}_1 \in \mathbb{R}^{C\times 2C}$ and $\mathbf{W}_2 \in \mathbb{R}^{C\times C}$. DWConv provides local bias, $\mathrm{SS2D}(\cdot)$ aggregates global context, and the gated residual stabilizes optimization [6], [2].

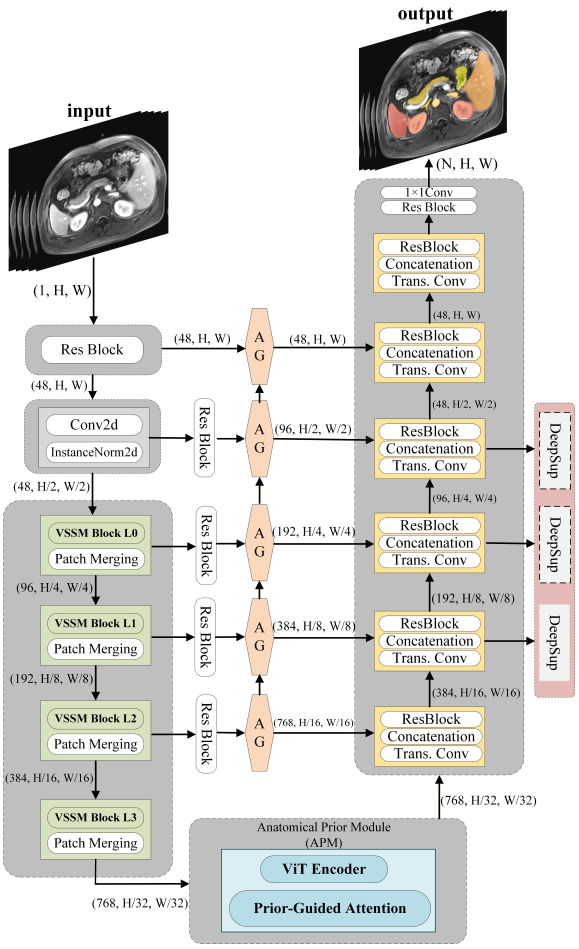


Fig. 1. The overall architecture of AP-Mamba. The network consists of a Mamba-based VSSM encoder, an Anatomical Prior Module (APM) applied at the bottleneck, and a structure-sensitive decoder with attention-gated skip connections for anatomy-aware feature reconstruction.

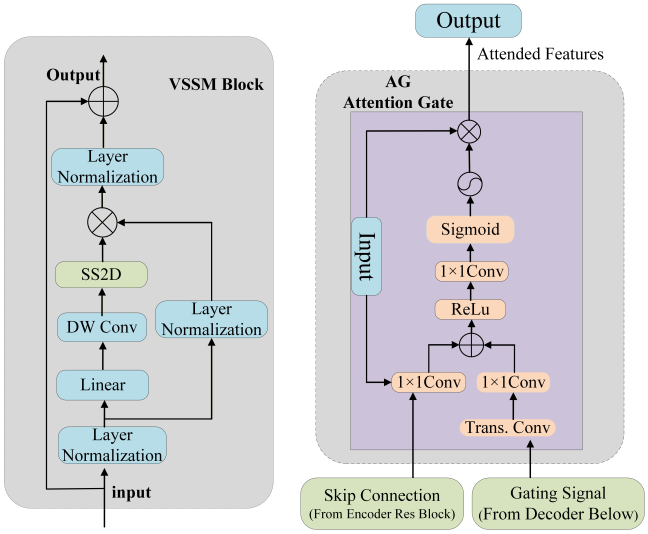


Fig. 2. Architectural details. (a) VSSM block. (b) Attention Gate (AG) used in the decoder.

B. Structure-Sensitive Decoder with Attention Gates (AG)

Attention Gates (AGs) are incorporated into the decoder to alleviate the semantic gap between encoder skip features and decoder representations [7]. Given a skip feature $\mathbf{x}$ and a gating signal $\mathbf{g}$, an additive gating function is adopted to compute attention coefficients that modulate $\mathbf{x}$ prior to skip fusion. *Attention Gate:* The gating signal is upsampled to the spatial size of $\mathbf{x}$:

$$\tilde{\mathbf{g}} = \mathcal{U}(\mathbf{g}; H, W), \tag{5}$$

$$\boldsymbol{\alpha} = \text{sigmoid}\Big(\psi\big(\text{ReLU}(\phi_x(\mathbf{x}) + \phi_g(\tilde{\mathbf{g}}))\big)\Big), \tag{6}$$

$$\hat{\mathbf{x}} = \mathbf{x} \odot \boldsymbol{\alpha}. \tag{7}$$

Here $\phi_x(\cdot)$ and $\phi_g(\cdot)$ are 1×1 linear projections that align channel dimensions, $\psi(\cdot)$ maps features to a single-channel attention logit, and $\odot$ denotes element-wise multiplication. The gated feature $\hat{\mathbf{x}}$ is then used for skip fusion in the decoder.

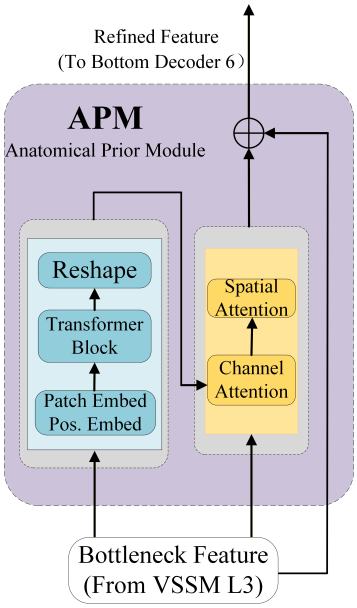


Fig. 3. Structure of the Anatomical Prior Module (APM).

C. Anatomical Prior Module (APM)

To capture global organ layout and co-occurrence patterns, we introduce an Anatomical Prior Module (APM) at the bottleneck. Given $X \in \mathbb{R}^{C\times H\times W}$, the APM consists of (i) an Anatomical Prior Encoder (APE) that extracts a global prior feature A, and (ii) a prior-guided attention (PGA) mechanism that modulates X with A (Fig. 3).

a) Anatomical Prior Encoder: APE is a lightweight ViT operating on bottleneck features: X is projected by a 1×1 conv, flattened into tokens with learnable 2-D positional embeddings (interpolated if needed), processed by several Transformer blocks, and projected back to obtain $A \in \mathbb{R}^{C\times H\times W}$:

$$A = \text{APE}(X). \tag{8}$$

b) Prior-guided modulation: We generate channel and spatial attention from A, denoted as $\mathcal{M}_c(A) \in \mathbb{R}^{C\times 1\times 1}$ (global pooling + MLP) and $\mathcal{M}_s(A) \in \mathbb{R}^{1\times H\times W}$ (1×1 conv). The output is computed by residual multiplicative modulation:

$$Y = X \odot \Big(1 + \lambda\, \mathcal{M}_c(A) \odot \mathcal{M}_s(A)\Big), \tag{9}$$

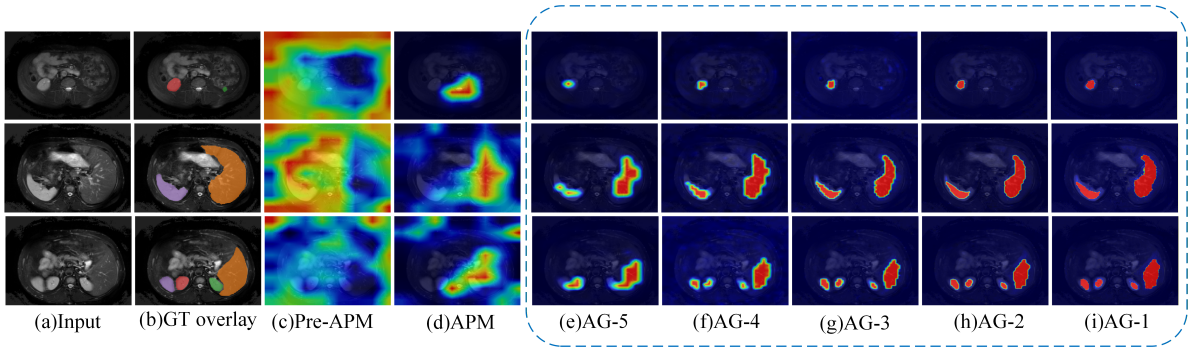


Fig. 4. Visualization of anatomical priors and attention gating in AP-Mamba. From left to right: (a) input MRI slice; (b) ground-truth overlay; (c) pre-APM channel-averaged bottleneck response before applying APM; (d) APM enhancement visualized as the channel-averaged magnitude $|F_{\text{APM}} - F_{\text{raw}}|$; (e–i) AG maps showing attention coefficients α from Attention Gates (AG-5 to AG-1, deep to shallow). All maps are min–max normalized to $[0, 1]$ and upsampled to the input resolution for visualization; warmer colors indicate higher responses/attention.

where λ is a learnable gain and $\odot$ denotes element-wise multiplication with broadcasting. Without PGA (ablation), we inject the prior by residual addition: $Y = X + \lambda A$.

D. Feature Response and Attention Map Visualization

To interpret how AP-Mamba exploits anatomical priors and attention gating, we visualize intermediate feature responses and gate coefficients in Fig. 4.

In column (c), the pre-APM bottleneck activation is spatially diffuse, indicating limited organ specificity. Column (d) shows the APM-induced feature change $|F_{\text{APM}} - F_{\text{raw}}|$, where responses concentrate around organ regions, suggesting improved coarse localization and suppression of spurious activations.

Columns (e)–(i) present attention-gate coefficients α from deep to shallow decoder stages (AG-5 to AG-1). Deeper gates highlight coarse regions, while shallower gates refine fine structures and boundaries. Overall, APM improves global localization consistency, and AG facilitates multi-scale refinement during decoding.

E. Ethics Statement

This study used only publicly available, de-identified datasets, including CHAOS and AMOS. No identifiable human subject data were used in this work; therefore, no additional institutional ethics approval was required for this study.

III. EXPERIMENTS

A. Datasets and Evaluation Protocol

The efficacy of AP-Mamba is evaluated on two public abdominal MRI benchmarks: CHAOS-MRI (T2-SPIR) and AMOS-MRI [8], [9]. For CHAOS-MRI, the test-set ground truth is not publicly available; therefore, 5-fold cross-validation is adopted to enable fair comparisons [8]. For AMOS-MRI, the official data split is followed and performance is reported on the held-out test set [9].

A 2D slice-based paradigm is used. Each 3D volume is sliced along the axial plane and treated as a collection of 2D training samples. During inference, slice-wise predictions are generated and then stacked to reconstruct the 3D volume for case-level evaluation.

B. Preprocessing and Data Augmentation

Data preprocessing followed the automated nnU-Net pipeline [10]. Raw MRI scans were cropped to non-zero regions to reduce redundancy. To handle resolution heterogeneity, images were resampled to a target spacing of 1.1875×1.1875 mm using third-order spline interpolation for MRI volumes and nearest-neighbor interpolation for masks to preserve label integrity. Intensities were then normalized by Z-score standardization ($\mu = 0, \sigma = 1$) using foreground-voxel statistics. Finally, the preprocessed volumes were extracted into 256×320 patches for training.

C. Implementation Details

AP-Mamba was implemented in PyTorch within the nnU-Net V2 framework and trained on NVIDIA A800 GPUs for up to 100 epochs with early stopping (patience 30). The AdamW optimizer was used with an initial learning rate of 1×10^{-4} and weight decay 5×10^{-2}, together with a cosine annealing scheduler ($\text{lr}_{\text{min}} = 1 \times 10^{-6}$). A parameter-specific learning-rate scheme was adopted, scaling the learning rates of the APM and Attention Gates by 2.0× and 1.5×, respectively. A warm-up stage was applied by freezing the encoder for the first 10 epochs. The objective combined Dice and Cross-Entropy losses, and deep supervision was enabled at four resolution scales.

D. Baselines and Evaluation Metrics

AP-Mamba is compared with representative state-of-the-art segmentation approaches spanning three architectural families:

- **CNN-based methods:** nnU-Net v2 [10], a widely adopted self-configuring baseline for medical image segmentation.
- **Transformer-based methods:** UNETR [11], Swin-UNETR [12], and nnFormer [13], which employ Transformer-style encoders to capture long-range dependencies [1].
- **Mamba-based methods:** U-Mamba (Bot/Enc) [14], SegMamba [15], and Swin-UMamba [16].

Segmentation accuracy is evaluated using the Dice similarity coefficient (DSC) and normalized surface distance (NSD), which respectively measure region overlap and boundary

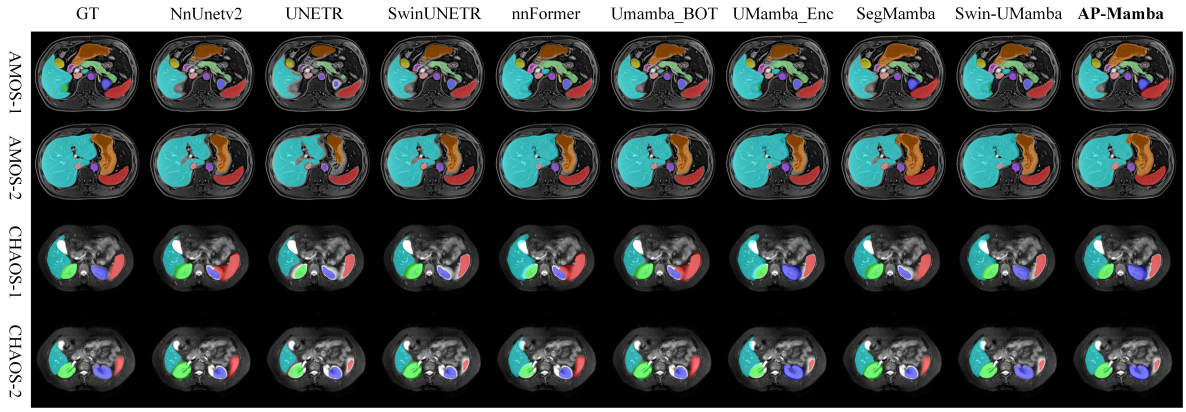


Fig. 5. Qualitative comparison of segmentation results on the AMOS-MRI and CHAOS-MRI datasets. Different colors denote different organ labels, with dataset-specific color mappings for AMOS-MRI and CHAOS-MRI. We compare AP-Mamba (last column) against representative state-of-the-art methods spanning CNN, Transformer, and Mamba architectures. Rows 1 and 2 correspond to different axial slices from the *same subject* in the AMOS dataset, while Rows 3 and 4 denote different slices from the *same subject* in the CHAOS dataset. It can be seen that AP-Mamba demonstrates superior anatomical fidelity, effectively reducing boundary leakage and preserving structural integrity compared to competing baselines.

agreement. Reported results are case-level averages: slice-wise predictions are stacked to reconstruct 3D volumes, and DSC/NSD are computed on the reconstructed volumes. In addition, the number of parameters is reported to reflect model scale.

E. Comparisons on Abdominal MRI Datasets

We compare AP-Mamba with representative CNN-, Transformer-, and Mamba-based methods, including nnU-Net [10], UNETR [11], Swin-UNETR [12], nnFormer [13], and recent Mamba variants (U-Mamba [14], SegMamba [15], Swin-UMamba [16]).

Table I summarizes results on CHAOS-MRI (5-fold CV) and AMOS-MRI (test set) [8], [9]. On CHAOS-MRI, AP-Mamba achieves 90.67% DSC and 91.10% NSD, outperforming Swin-UMamba (90.52% DSC) and nnU-Net (85.45% DSC). On AMOS-MRI, AP-Mamba attains the best performance (82.32% DSC, 81.84% NSD), improving over Swin-UMamba by +0.94% DSC and over nnU-Net by +3.63% DSC. These results indicate that incorporating explicit anatomical priors improves robustness under complex anatomical variation.

F. Qualitative Comparison

To further assess perceptual segmentation quality, Fig. 5 shows a qualitative comparison with representative methods. For both AMOS and CHAOS, two axial slices from the same subject are visualized to examine within-subject consistency across anatomical sections.

In AMOS (Rows 1–2), low-contrast organs (e.g., stomach and spleen) often challenge Transformers (UNETR [11], Swin-UNETR [12]), which may exhibit boundary artifacts or over-segmentation. Recent Mamba-based models (SegMamba [15], Swin-UMamba [16]) generally improve global coherence but can still miss fine details in complex multi-organ scenes.

AP-Mamba yields segmentations that align more closely with the ground truth, showing fewer false positives and reduced boundary leakage. This improvement is attributed to the global guidance from APM and the local refinement from AG. Similar gains are observed on CHAOS (Rows 3–4), where adjacent organs are separated with sharper boundaries across different slices of the same patient.

TABLE I

QUANTITATIVE COMPARISON ON CHAOS-MRI AND AMOS-MRI DATASETS. METRICS: MEAN DSC (%) AND MEAN NSD (%). THE BEST RESULTS ARE HIGHLIGHTED IN **BOLD**, AND THE SECOND-BEST ARE UNDERLINED.

Method	Params (M)	CHAOS (5-fold)		AMOS (Test)	
		DSC (↑)	NSD (↑)	DSC (↑)	NSD (↑)
CNN-based Methods					
nnU-Net v2 [10]	45.0	85.45	85.57	78.69	76.05
Transformer-based Methods					
UNETR [11]	92.8	71.43	70.23	70.20	61.26
Swin-UNETR [12]	25.4	75.43	74.60	76.88	72.87
nnFormer [13]	58.6	76.11	75.35	77.10	73.26
Mamba-based Methods					
U-Mamba Bot [14]	24.4	88.58	88.99	80.05	78.40
U-Mamba Enc [14]	64.4	89.18	89.65	81.10	80.23
SegMamba [15]	66.9	90.09	90.64	79.41	77.32
Swin-UMamba [16]	62.2	<u>90.52</u>	<u>90.76</u>	<u>81.38</u>	<u>80.72</u>
AP-Mamba (Ours)	69.0	**90.67**	**91.10**	**82.32**	**81.84**

G. Ablation Studies

To investigate the contribution of each component, we conduct ablation studies on both CHAOS-MRI and AMOS-MRI, using Swin-UMamba [16] as the baseline. We incrementally add the Anatomical Prior Module (APM) and Attention Gates (AG), and report the Dice similarity coefficient (DSC) and normalized surface distance (NSD). The results are summarized in Table II.

Adding APM consistently improves the baseline on both datasets (e.g., +0.37% DSC on AMOS), indicating that explicitly modeling global anatomical context provides effective guidance under ambiguous conditions. By capturing organ co-occurrence and spatial layout priors, APM stabilizes localization and reduces anatomically implausible activations, particularly in low-contrast regions.

TABLE II

Ablation study of the Anatomical Prior Module (APM) and Attention Gates (AG) on CHAOS-MRI (5-fold CV) and AMOS-MRI (held-out test set). Metrics: mean DSC (%) and mean NSD (%). Best in **bold** and second-best underlined.

Method	Modules		CHAOS-MRI		AMOS-MRI	
	AG	APM	DSC (↑)	NSD (↑)	DSC (↑)	NSD (↑)
Swin-UMamba (Baseline)			90.52	90.76	81.38	80.72
+ AG only	✓		89.69	90.32	81.16	79.29
+ APM only		✓	**90.81**	**91.85**	81.75	80.59
AP-Mamba (Full)	✓	✓	90.67	91.10	**82.32**	**81.84**

In contrast, using AG alone results in a slight performance drop, suggesting that purely local gating without global calibration is susceptible to background noise and may suppress informative cues or amplify spurious responses in challenging areas.

When AG is combined with APM, the full AP-Mamba achieves the best performance on AMOS-MRI (DSC 82.32%, NSD 81.84%). This supports a complementary interpretation: the global prior from APM calibrates the gating signal, enabling AG to suppress noise more reliably while preserving organ-specific structures. The synergy improves boundary refinement and enhances generalization on the more diverse AMOS benchmark.

On CHAOS-MRI, the APM-only variant performs best. This may reflect dataset-dependent characteristics: for smaller benchmarks, global regularization alone can be sufficient, whereas additional local gating becomes more beneficial as anatomical variability and scene complexity increase, as observed in AMOS-MRI.

IV. CONCLUSION

In this work, we present AP-Mamba, a multi-organ abdominal MRI segmentation framework that combines the linear-complexity efficiency of state space models [2] with explicit anatomical priors. Specifically, we propose a lightweight Transformer-based Anatomical Prior Module (APM) to capture global organ layout and co-occurrence cues [1], and a structure-sensitive decoder with attention-gated skip connections to suppress background interference and refine organ boundaries. Experiments on CHAOS-MRI and AMOS-MRI consistently improve both region overlap and boundary accuracy over strong baselines. In future work, we will extend AP-Mamba to fully 3D volumetric processing and further evaluate its generalization to other modalities such as CT; the proposed prior modeling strategy may also benefit natural image segmentation and broader vision tasks.

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